



Read and Change a Loop

Quick facilitation notes & answer key

Goal

Students read an existing loop to state what it does, then alter the repeat count or the inside block and describe how the outcome changes (Ontario C3.2). Includes a work-backwards challenge.

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

A block-coding app (Scratch-style) to project; the printed worksheet + a pencil per student.

Run it (about 30 min)

1. Hook: read a recipe before changing an amount — read code before you change it.
2. Project the repeat-5 move-10 loop; read it together as 5 jumps of 10 = 50.
3. Students do Exercises 1–4 (Ex 4 is the stretch), solo or in pairs.
4. Share answers; land the big idea: count changes how many; inside changes what.

Answer key (checked by the run-gate)

Ex 1 — 5 moves; 50 steps total.

Ex 2 — 3 moves; 30 steps.

Ex 3 — 10 steps (5×2); changing the inside block changed how far each move goes.

Ex 4 (challenge) — repeat 4 ($40 \div 10 = 4$).

Watch for & differentiate

Common slip: changing the count when the task asks to change the inside block (Ex 3), or vice versa.

Support: have students underline the part the question changes before solving.

Extend: ask what repeat gives 60 steps at 10 each (6).

Success looks like

The student reads a loop to state its result (Ex 1), changes the count and predicts (Ex 2), changes the inside block and describes the effect (Ex 3), and works backwards to find a needed count (Ex 4).